

OPTIMIZATION TECHNIQUE QUESTION BANK

A.	Explain the concept of classical optimization methods.	
B.	Using the method of lagrange's multiplier, solve the following NLPP. Optimize $Z = 6x_1^2 + 5x_2^2$ Subject to $x_1 + 5x_2 = 7$ $x_1, x_2 \geq 0$	
C.	Solve the following NLPP by Kuhn Tucker condition: Maximize $Z = 10x_1 + 4x_2 - 2x_1^2 - x_2^2$ Subject to $2x_1 + x_2 \leq 5$ $x_1, x_2 \geq 0$	
D.	Explain the concept of an optimization technique. A. What are the components of an optimization problem? B. Using the method of lagrange's multiplier, solve the following NLPP. Optimize $Z = 4x_1 + 8x_2 - x_1^2 - x_2^2$ Subject to $x_1 + x_2 = 4$ $x_1, x_2 \geq 0$ C. Solve the following NLPP by Kuhn Tucker condition: Maximize $Z = 2x_1^2 - 7x_2^2 + 12x_1x_2$ Subject to $2x_1 + 5x_2 \leq 98$ $x_1, x_2 \geq 0$ D. What is the application of an optimization technique?	
A.	Explain linear programming problems with a suitable example.	
B.	Solve the following LPP by simplex method Maximize $Z = 3x_1 + 5x_2$ Subject to $3x_1 + 2x_2 \leq 18$; $0 \leq x_1 \leq 4$; $0 \leq x_2 \leq 6$; $x_1, x_2 \geq 0$	
C.	Using penalty (Big-M or Charne's) method solve the following LPP Maximize $Z = 3x_1 - x_2$ Subject to $2x_1 + x_2 \geq 2$; $x_1 + 3x_2 \leq 3$; $x_2 \leq 4$; $x_1, x_2 \geq 0$	
D.	What are the characteristics of integer linear programming ?	
	A. What are the steps required to solve the LPP by simplex method? B. Solve the following LPP by simplex method. Minimise $Z = x_1 - 3x_2 + x_3$ Subject to $3x_1 - x_2 + 2x_3 \leq 7$;	

	$2x_1 + 4x_2 \geq -12;$ $-4x_1 + 3x_2 + 8x_3 \leq 10; x_1, x_2, x_3 \geq 0$ C. Using penalty (Big-M or Charne's) method solve the following LPP Maximize $Z = 3x_1 - x_2$ Subject to $2x_1 + x_2 \leq 2;$ $x_1 + 3x_2 \geq 3;$ $x_2 \leq 4; x_1, x_2 \geq 0$ D. Explain the Two-Phase Method used to solve Linear Programming Problems (LPP) with artificial variables. Clearly describe the steps involved in Phase I and Phase II of the method. Illustrate your answer with a suitable example.	
A.	Use the Bisection Method to find a root of the equation. $f(x) = x^3 - x - 2$ correct up to 3 decimal places, within the interval $[1,2]$. Show all steps.	
B.	Find the real root of the equation $x^4 - x - 10 = 0$ correct to three decimal places by newton raphson method where $x_0 = 1.6$	
C.	Explain the procedure of the Bisection Method step by step.	
D.	Use the Secant Method to find a root of the equation. $f(x) = x^3 - x - 2$ starting with $x_0 = 1$ and $x_1 = 2$ Perform iterations until the difference between successive approximations is less than 10^{-4} Show all steps and give the approximate root.	
A.	Explain the Bisection Method used for finding the roots of nonlinear equations. Write down the step-by-step procedure of the method?	
B.	Find the real root of the equation $3x - \cos x - 1 = 0$ correct to three decimal places by newton raphson method where $x_0 = 0.6$	
C.	Use the Secant Method to find a root of the equation. $f(x) = x^3 + x^2 - 3x - 3$ starting with $x_0 = 1$ and $x_1 = 2$ Perform iterations until the difference between successive approximations is less than 10^{-4} Show all steps and give the approximate root.	
D.	Use the Bisection Method to find a root of the equation. $f(x) = x^2 - 3$ correct up to 3 decimal places, within the interval $[1,2]$. Show all steps.	
A.	Explain Simulated Annealing Algorithm.	
B.	What are the common crossover techniques used in Genetic Algorithms? Explain each technique with the help of suitable examples.	
C.	Use the Golden Section Search Method to minimize the function $f(x) = (x - 2)^2$ within the interval $[0,4]$. Show all the steps involved in the solution.	
D.	What are the advantages and disadvantages of particle swarm optimization? A. Explain the concept of Particle Swarm Optimization (PSO) .	

	B. Explain the concept of mutation with an example. C. Use the Golden Section Search Method to maximize the function $f(x) = 4\sin\theta(1 + \cos\theta)$ within the interval $[0, \pi]$ Show all the steps involved in the solution. D. Differentiate between genetic algorithm and genetic programming.
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What are the applications of lagrange's multiplier?
Explain working procedure for dual simplex method.
Explain the bisection method with a suitable example.
Differentiate between genetic algorithm and genetic programming.
Explain the types of an optimization problem. Write down the comparison between regular simplex method and dual simplex method? What are the steps required to solve the bisection method? What are the different types of reproduction operators used in Genetic Algorithms? Explain each with suitable examples.